

GABRIEL NIXON RAJ

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EDUCATION	Center for Data Science, New York University <i>M.S. in Data Science</i> • Current GPA: 3.8/4.0	New York, NY, USA 2024 – 2026 (<i>expected</i>)
	SIES Graduate School of Technology, University of Mumbai <i>B.E. in Computer Engineering</i> • GPA: 3.7/4.0	Mumbai, India 2019 – 2023
RESEARCH INTERESTS	Q1) My research focuses on how agents adapt when even their underlying assumptions may be wrong. Q2) I also study how systems should behave when coordination, rather than computation, becomes the primary bottleneck.	
RESEARCH EXPERIENCE	Northeastern University (Gyori Lab) Capstone Researcher	Sep 2025 – Present
	• Problem: Biomedical knowledge bases are fragmented, making it hard to link. • Approach: Applied logic-based tools (LogMap) and deep learning methods (BERTMap) for ontology alignment. • Impact: Improved interoperability of biomedical data; preparing a manuscript on scalable ontology integration.	
	CUNY Geospatial Institute Associate Researcher	Aug 2025 – Present
	• Problem: Urban decision-making often relies on fragmented satellite data. • Approach: Developed optimization-driven deep learning algorithms. • Impact: Improved efficiency in geospatial analysis for city planning; co-authoring two manuscripts on geospatial AI and decision support systems.	
	NYU Mattar Lab Reinforcement Learning Researcher	Aug 2025 – Present
	• Problem: In dynamic environments, agents often need to adapt when conditions change unpredictably. • Approach: Modeled adaptive strategies using recurrent neural networks (RNNs) and hidden Markov models (HMMs). • Impact: Framed adaptive policy selection as an optimization problem; work is contributing to a forthcoming reinforcement learning manuscript.	
	NYU School of Global Public Health AI Policy Researcher	Jun 2025 – Sep 2025
	• Problem: AI systems in public health can amplify bias and fail under uncertainty. • Approach: Analyzed trust, bias, and coordination risks through policy modeling. • Impact: Produced equity-aware ML deployment recommendations.	
	Lawrence Berkeley National Laboratory AI Research Intern	May 2025 – Aug 2025
	• Problem: HPC systems like Perlmutter generate massive telemetry streams, making real-time monitoring difficult. • Approach: Designed a multi-agent decision-support framework with routing mechanisms to coordinate diagnostics across heterogeneous data streams. • Impact: Enabled real-time summarization and improved reliability of HPC infrastructure; presented at Berkeley Lab CS Program.	
	SIES Graduate School of Technology Research Assistant	May 2023 – Aug 2023
	• Problem: Optimization algorithms struggle with local minima in search problems. • Approach: Developed hybrid genetic-annealing algorithms, tuning heuristics to balance exploration and exploitation. • Impact: Stabilized optimization performance under structural noise; led to a technical report under Prof. Anindita Khade.	

PUBLICATIONS & RESEARCH OUTPUTS

Conferences

1. **G. N. Raj.** *Belief Distortion under Stress using Trust Decay and KL-Regularized Inference.* Oral Presentation at CFE-CMStatistics 2025, University of London.
2. **G. N. Raj.** *Stress-Aware Scenario Generation for Reliable Portfolio Inference under Regime Shifts.* NeurIPS 2025 GenAI in Finance.
3. **G. N. Raj, Ramesh T., Samuel S.** *YouTube Transcript Summarization using Transformer-Based Models.* Oral presentation at ICREACT 2023.
4. **A. T. Nadar, S. Chandane, G. N. Raj, N. M. Pasi, Y. A. Patil.** *Automated Energy Billing with Blockchain and the Prophet Forecasting Model.* Presented at IEEE MRT 2023.

Journals & Book Chapters

1. Nishant Kathpalia, **G. N. Raj**, Madhavan Venkatesh. *A Smart Healthcare Companion with Tesseract OCR and KNN Integration.* Technologies for Energy, Agriculture, and Healthcare (CRC Press, 2025, Chapter)
2. **A. T. Nadar, G. N. Raj, S. Chandane, S. Bhat.** *Leveraging CNN and IoT for Effective E-Waste Detection in Smart Cities.* IEEE I-SMAC 2023 / IEEE Xplore

Posters

1. **G. N. Raj, S. Zhang, T. Zhou, A. Allahverdiyeva.** *Sharp Minima Can Generalize For Deep Nets.* Poster presented at NYU Center for Data Science, 2025.
2. **G. N. Raj, K. Konate, W. Bhimji.** *Monitoring Meets Modularity: A Framework for Scalable HPC Telemetry Across Diverse Sources.* Poster, Lawrence Berkeley National Laboratory.

INTERNSHIPS & APPLIED EXPERIENCE

- NYU Entrepreneurial Institute** | New York, NY 2025.01 – 2025.05
- Built a RAG agent for multimodal document search.
 - Used Whisper ASR, PyMuPDF, BM25, and OpenAI embeddings over FAISS/Pinecone.
 - Resulted in higher retrieval precision and reduced query latency.
- Impact Global Digital** | Mumbai, India 2023.08 – 2024.04
- Deployed a neural document search system for healthcare logistics.
 - Applied CNN/RNN embeddings on AWS with TF-IDF, BM25, and FAISS retrieval.
 - Improved robustness of search and supported medical data workflows.
- The Tee Company (Co-Founder)** | Mumbai, India 2022.03 – 2022.12
- Created a pricing and inventory optimization model for retail operations.
 - Implemented adaptive decision-making with bipartite recommendation matching.
 - Boosted sales efficiency and optimized stock turnover.
- Private Investor Research Group** | Mumbai, India 2022.03 – 2023.02
- Built a regime forecasting pipeline on 50+ macro-financial indicators.
 - Used ensemble models to design a probabilistic entry–exit signal framework.
 - Helped optimize portfolio allocation under regime uncertainty.

TEACHING EXPERIENCE	CUNY Geospatial Institute New York, USA 2025
	- Designed and delivered tutorials on Geospatial AI for a graduate-level class. - Focused on spatial data handling, visualization, and integration with machine learning models.
	University of Munich Munich, Germany Summer 2025
	- Taught Operations Research in both German and English. - Led seminar-style sessions with 5–6 students, emphasizing applied optimization.
HONORS & LEADERSHIP	SIES Graduate School of Technology Mumbai, India 2022 – 2023
	- Teaching Assistant for Probabilistic Modeling. - Conducted practice and review sessions for 60+ undergraduate students.
	Father Agnel's Ashram Mumbai, India 2018 – 2020
	- Volunteer educator for mathematics and basic quantitative reasoning. - Taught over 30 underprivileged children in small classroom settings.
SKILLS	Paragon Fellowship (AI policy and election systems). Gold Medalist, National Math Olympiad (Saudi Arabia). Winner, CSI International Project Competition 2023. Abacus: Completed all 8 levels; represented Saudi Arabia internationally. Leadership: NYU Graduate Student Council, Social Inclusion Board, SIES AI Council.
	Languages : Python, R, SQL.
	Math/Optimization : Linear algebra, probability, stochastic processes, convex optimization, dynamic programming, game theory, MDPs, Bayesian inference, heuristic/metaheuristic methods; optimization solvers (CVXPY, Gurobi, CPLEX).
	Libraries/Frameworks : NumPy, Pandas, TensorFlow, PyTorch, Flask, Django.
ACADEMIC SERVICES	Big Data : Apache Kafka, Hive, MapReduce, AWS, Azure, Delta Lakes.
	Databases : MySQL, PostgreSQL, SQLite, MongoDB, Snowflake.
	Tools : Git, VS Code, Jupyter, PyCharm, RStudio, Google Colab.
	Reviewer for: <i>NeurIPS 2025 Education Program</i>